University of Tabuk
Faculty of Computers & Information Technology
Department of Computer Engineering

MICROPROCESSORS & EMBEDDED SYSTEMS LAB
CEN-412

MICROPROCESSORS & EMBEDDED SYSTEMS LAB CEN-412

Project (5). Home controllable humidifier 20 points

1. Project description:

This project is a Home Humidifier Control System built using Arduino.
The system monitors the surrounding humidity and temperature using a DHT11 sensor, and it controls a mist maker and a DC fan to maintain a comfortable humidity level inside a room.

A potentiometer is used to set the desired humidity level.
When the current humidity is lower than the desired level, the system activates the Mist Maker (via a relay) to increase humidity.
When the humidity reaches or exceeds the desired value, the Mist Maker automatically turns OFF.

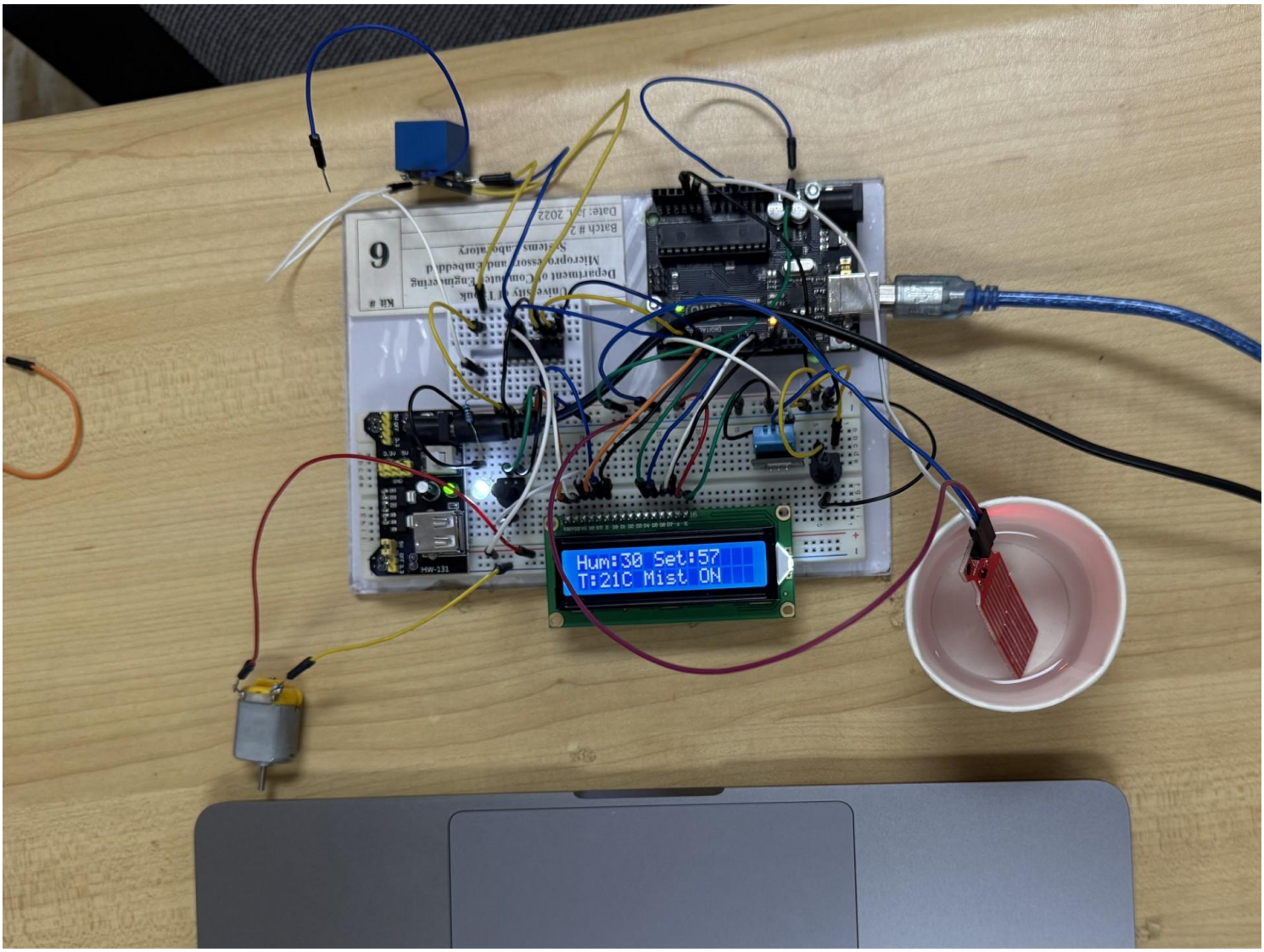
A water-level sensor ensures safe operation. If the water level is too low, the system disables all outputs (mist + fan) and displays a warning message.

A DC fan, controlled through the L293D motor driver, turns ON and OFF together with the mist maker. A visual indicator LED shows the status of the mist maker. An LCD 16x2 display provides real-time humidity, temperature, and system status.

This project demonstrates real-time monitoring and environmental control, combining sensors, actuation, and display modules in an automated system.

2. Project schematic Diagram

In lab:



3. Algorithm and Program source code:

Algorithm :

Start

Initialize LCD, DHT11 sensor, relay, LED, fan (L293D), and inputs.
Display "System Ready" on the LCD.

Loop:

1. Read humidity and temperature from DHT11.
2. Read the water level from water sensor.
3. Read desired humidity using potentiometer.
4. If water level is LOW:
    - Turn OFF Mist Maker
    - Turn OFF Fan
    - Turn OFF LED
    - Display "Water LOW" message
    - Skip to next loop cycle

5. Otherwise (water OK):
    - Compare humidityNow with humiditySet

    IF humidityNow < humiditySet:
        - Turn ON Mist Maker
        - Turn ON LED
        - Turn ON Fan

    ELSE (humidityNow >= humiditySet):
        - Turn OFF Mist Maker
        - Turn OFF LED
        - Turn OFF Fan

6. Display:
    - Current humidity
    - Set humidity
    - Temperature
    - Mist status (ON/OFF)

7. Wait 700 ms
Repeat loop

Program Source Code:

```
#include <SimpleDHT.h>
#include <LiquidCrystal.h>

LiquidCrystal lcd(7, 8, 9, 10, 11, 12);

int pinDHT = 2;
SimpleDHT11 dht11;

int waterPin = A0;
int potPin   = A1;

int relayPin   = 6;
int mistLedPin = 13

#define ENABLE 5
#define DIRA   3
#define DIRB   4

void setup() {
  Serial.begin(9600);

  lcd.begin(16, 2);
  lcd.print("Humidifier Ready");

  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, HIGH);
  pinMode(mistLedPin, OUTPUT);
  digitalWrite(mistLedPin, LOW);

  pinMode(ENABLE, OUTPUT);
  pinMode(DIRA, OUTPUT);
  pinMode(DIRB, OUTPUT);

  digitalWrite(DIRA, LOW);
  digitalWrite(DIRB, LOW);
  digitalWrite(ENABLE, LOW);

  delay(1500);
}

void loop() {

  // Read DHT11
  byte temperature = 0;
  byte humidityNow = 0;
  dht11.read(pinDHT, &temperature, &humidityNow, NULL);
```

```
int humiditySet = map(analogRead(potPin), 0, 1023, 30, 80);

int waterValue = analogRead(waterPin);
bool waterLow = (waterValue < 50);

if (waterLow) {
  digitalWrite(relayPin, HIGH);
  digitalWrite(mistLedPin, LOW);

  digitalWrite(ENABLE, LOW);
  digitalWrite(DIRA, LOW);
  digitalWrite(DIRB, LOW);

  lcd.clear();
  lcd.print("Water Level LOW");
  lcd.setCursor(0, 1);
  lcd.print("Fill tank!");
  delay(700);
  return;
}


lcd.clear();
lcd.print("Hum:");
lcd.print(humidityNow);
lcd.print(" Set:");
lcd.print(humiditySet);

lcd.setCursor(0, 1);
lcd.print("T:");
lcd.print(temperature);
lcd.print("C ");

if (humidityNow < humiditySet) {
  // Mist ON
  digitalWrite(relayPin, LOW);
  digitalWrite(mistLedPin, HIGH);

  digitalWrite(ENABLE, HIGH);
  digitalWrite(DIRA, LOW);
  digitalWrite(DIRB, HIGH);

  lcd.print("Mist ON");
} else {
  // Mist OFF
  digitalWrite(relayPin, HIGH);
  digitalWrite(mistLedPin, LOW);
```

```
    digitalWrite(ENABLE, LOW);
    digitalWrite(DIRA, LOW);
    digitalWrite(DIRB, LOW);

    lcd.print("Mist OFF");
   }

  delay(700);
}
```

4. Program description:

This program implements a humidity control system using Arduino.
The DHT11 sensor measures the current **humidity and temperature**, while a potentiometer determines the **desired humidity level**.

The system first checks the water level using a water sensor. If the water is too low, it disables all operations and shows a warning on the LCD to protect the Mist Maker.

When water is available:

- If the measured humidity is **below** the user-set humidity:
    - The **Mist Maker** turns ON through a relay.
    - The **indicator LED** lights up.
    - The **DC Fan** turns ON (via L293D).
- If the humidity reaches/exceeds the set value:
    - Mist Maker turns OFF.
    - LED turns OFF.
    - Fan turns OFF.

The LCD displays:

- Current humidity
- Desired humidity (Set value)
- Temperature
- Mist ON/OFF status

This system provides a simple, safe, and automated method of controlling indoor humidity using sensors, relay control, motor driver, and user input.